

1. (Fall 2024, #4) Consider the region bounded by $y = x^2 - x^3$, and $y = 0$, where $x \geq 0$. Find an integral formula for the volume of the solid obtained

(a) if we rotate the region about the x -axis (do not calculate the actual value),

(b) and if we rotate the region about the y -axis (do not calculate the actual value).

2. (Fall 2023, #4)

- (a) Let $\mathcal{R}$ be the region bounded by the x -axis and the curve $y = e^x + x$, where $0 \leq x \leq 2$. Compute the volume of the solid obtained by rotating $\mathcal{R}$ about the y -axis.

- (b) The region $\mathcal{R}$ is bounded by the curves $x = 0$, $y = x^4 + x^2$, and $y = x^2 + 16$. Set up an integral expressing the volume of the solid obtained by rotating $\mathcal{R}$ about $y = -1$. **You do not need to evaluate the integral.**

3. (Spring 2023, #2) Find the volume of the solid obtained by rotation of the region bounded by the curves

$$y = \frac{e^{x/3}}{2-x}, \quad x = 1, \quad x = 2, \quad y = 0$$

about the line $x = 2$. *Warning: you may obtain an improper integral.*

4. (Fall 2022, #5) Let $\mathcal{R}$ be the region containing the point $(3, 6)$ enclosed by the curves $y = 6x - x^2$, $y = x$, and $y = 18 - 5x$. Set up, but **do not evaluate**, an integral expressing the **volume** of the solid obtained by rotating $\mathcal{R}$ about $x = 1$.

5. (Fall 2021, #5) Consider the region R bounded by $y = x^2 - 6$ and $y = -x^2 + 16x + 12$. Set up, but DO NOT EVALUATE, an integral for the volume of the solid obtained by rotating the region R :

(a) about the line $y = -6$,

(b) and about the line $x = -6$.